NAMAA at ArGuard Shared Tasks: Multimodal Fusion, Vision–Language Model Adaptation, and In-Context Prompting for Arabic Hateful Meme Detection

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Abstract

Multimodal hate speech detection requires interpreting how an image and its accompanying text jointly construct a target, stance, or derogatory implication. In Arabic memes, this relation is often shaped by locally recognizable public figures, political references, visual analogies, and indirect humor, making either modality insufficient on its own. We tackle this problem in Track A of ArGuard 2026, covering binary hateful meme detection (A1) and fine-grained multi-label classification (A2). Our experiments compare jointly trained dual encoders, parameter-efficient vision–language model (VLM) adaptation, and frozen-model prompting. We fine-tune Qwen3-VL, Llama 3.2 Vision, and Gemma 4 with QLoRA using task-specific binary and multi-label objectives. We also build CLIP–MARBERTv2 and SigLIP–AraBERT-Twitter cross-attention systems and evaluate frozen VLMs with zero-shot prompting and in-context learning. Qwen3-VL achieves our highest A1 score (0.7610), while SigLIP–AraBERT-Twitter leads A2 (0.3851), where label sparsity and multi-label prediction remain challenging.

1 Introduction

Memes generally communicate through the interaction of an image and a short textual message, often relying on juxtaposition, visual analogy, and an implied target. In hateful memes, neither modality need be explicitly harmful on its own, because the derogatory meaning may emerge only when both are read together (Kiela et al., 2020).

Arabic meme interpretation also depends heavily on locally recognizable figures, political events, and culturally situated humor, while stylized text and imperfect OCR can obscure the linguistic signal. Recent datasets and shared tasks have begun to make this setting measurable, moving from Arabic propaganda memes toward multimodal hate detection and fine-grained hate annotation (Alam et al., 2024, 2025; Zaghouani et al., 2025; Kmainasi et al., 2026).

We study this problem through Track A of ArGuard 2026 (Alam et al., 2026), which covers binary hateful meme detection (A1) and multi-label prediction of fine-grained hate categories (A2). Our central question is how different forms of task adaptation behave when only a few thousand labeled memes are available. We fine-tune instruction-tuned VLMs to generate task-label sequences and compare them with discriminative dual encoders that align Arabic token representations with image patches through bidirectional cross-attention (Vaswani et al., 2017). For A1, we also evaluate frozen VLMs with zero-shot prompting and in-context learning (ICL), including a retrieval-augmented configuration.

Qwen3-VL attains the highest score in subtask A1 with a macro-F_1 of 0.7610, while the SigLIP-based dual encoder scores the best in subtask A2 at 0.3851. The dual-encoder soft vote also remains competitive on A1, with a macro-F_1 of 0.7443. Among the exploratory prompting baselines, static multimodal ICL scores highest, ahead of both zero-shot and retrieval-based ICL. Beyond lexical cues, our error analysis highlights challenges with targets, stance, cultural context, and nonliteral humor.

2 Related Work

The Hateful Memes Challenge established that multimodal hate detection cannot be reduced reliably to unimodal shortcuts (Kiela et al., 2020). Research on Arabic memes has progressed from ArMeme's propaganda annotations (Alam et al., 2024) to the coarse- and fine-grained hate labels in Prop2Hate-Meme (Alam et al., 2025) and the OCR-aware hateful meme task introduced by MAHED (Zaghouani et al., 2025). Recent benchmarks broaden this scope: AraHarMeme targets harmful Arabic memes, while AHA-Memes provides

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fine-grained multi-label hate annotations and baselines spanning unimodal encoders, fusion, prompting, and VLM fine-tuning (El-Amrany et al., 2026; Kmainasi et al., 2026). MemeIntel studies explanation generation for Arabic propaganda memes and English hateful memes (Kmainasi et al., 2025).

Arabic-specific encoders remain important for social-media variation, and the BERT-based Arabic encoder family continues to be extended, most recently by AraModernBERT (Elshehy et al., 2026). We use MARBERTv2 (Abdul-Mageed et al., 2021) and AraBERT-Twitter (Antoun et al., 2020) for their coverage of informal and dialectal Arabic, pairing them with CLIP (Radford et al., 2021) and SigLIP (Zhai et al., 2023). Bidirectional cross-attention enables exchange between token and patch sequences (Vaswani et al., 2017). In a related text-only setting, Djamai et al. (2026) compare fine-tuned encoders with zero-shot prompting and retrieval-augmented ICL for Arabic polarization detection and type classification.

Instruction-tuned VLMs offer a different route by treating classification as label-sequence generation. We adapt Qwen3-VL (Bai et al., 2025), Llama 3.2 Vision (Meta AI, 2024), and Gemma 4 (Gemma Team, 2026) through parameter-efficient fine-tuning, and evaluate Gemma 4 31B through frozen-model prompting, both in zero-shot and ICL settings, including a retrieval-based variant.

3 Task and Evaluation

Each Track A example consists of a meme image and its OCR text (Alam et al., 2026; Kmainasi et al., 2026). Subtask A1 requires a binary prediction of *Hateful* or *Not Hateful*, whereas subtask A2 assigns all applicable labels from the ten categories with training-set support: *Contempt*, *Dehumanization*, *Exclusion*, *Humor*, *Incitement*, *Inferiority*, *Mocking*, *Other*, *Sarcasm*, and *Slurs*. Five further taxonomy classes (*Extremism*, *Historical*, *Insults*, *Stereotyping*, *Threat*) have no support in the released data and are therefore never predicted. The A2 label distribution is long-tailed, and some categories—particularly *Exclusion*—have very few training-set examples.

The released dataset contains 3,500 examples in the training split and 500 examples in each of the development split, development-test split, and final test split (Alam et al., 2026; Kmainasi et al., 2026). We use the development split for model selection and threshold tuning. Macro-F_1 is the primary metric for both subtasks.

4 Approaches

We investigate two families of trained systems together with a set of exploratory frozen-model prompting baselines (for subtask A1 only). Figure 1 gives an overview of the three paradigms.

4.1 Fine-tuned generative VLMs

We represent each example as a multimodal chat turn containing the meme image, OCR text, and task instruction. Using Unsloth, we fine-tune Qwen3-VL-8B-Instruct, Llama 3.2 Vision 11B, and Gemma 4 E4B with 4-bit QLoRA (Bai et al., 2025; Meta AI, 2024; Gemma Team, 2026; Dettmers et al., 2023; Han et al., 2023). We apply adapters to both the vision and language components, including their attention and feed-forward projections.

Task formulations. For an input $x = (I, o, q)$ containing image I, OCR text o, and task instruction q, let $y_{1:T}$ be the target label sequence. The trainable adapters minimize token-level negative log-likelihood, $\mathcal{L}_{\text{gen}} = -\sum_t \log p_\theta(y_t \mid I, o, q, y_{<t})$. We train a separate model for each subtask. For A1, the target y is either `Hateful` or `Not Hateful`; for A2, y serializes all applicable categories as a comma-separated sequence. We use deterministic decoding and post-process the generated outputs by matching them to the official label vocabulary, removing duplicates, and placing the labels in a canonical order. Appendix B.1 provides the exact instructions.

4.2 Dual-encoder fusion and ensembling

Architecture and objective. Figure 1(b) illustrates the architecture. Text tokens and image patches are linearly projected and layer-normalized into a shared d=512-dimensional space, giving token matrix $T \in \mathbb{R}^{n\times d}$ and patch matrix $V \in \mathbb{R}^{m\times d}$. Suppressing residual and feed-forward notation, bidirectional cross-attention and pooling are written as

$$\widetilde{T} = \mathrm{MHA}(T, V, V), \qquad \widetilde{V} = \mathrm{MHA}(V, T, T),$$
$$h = [\mathrm{MP}(\widetilde{T}); \mathrm{AP}(\widetilde{V})].$$

Here, MP denotes masked mean pooling over text tokens, while AP denotes average pooling over image patches. Padded text positions are also masked in attention. Each block adds residual,

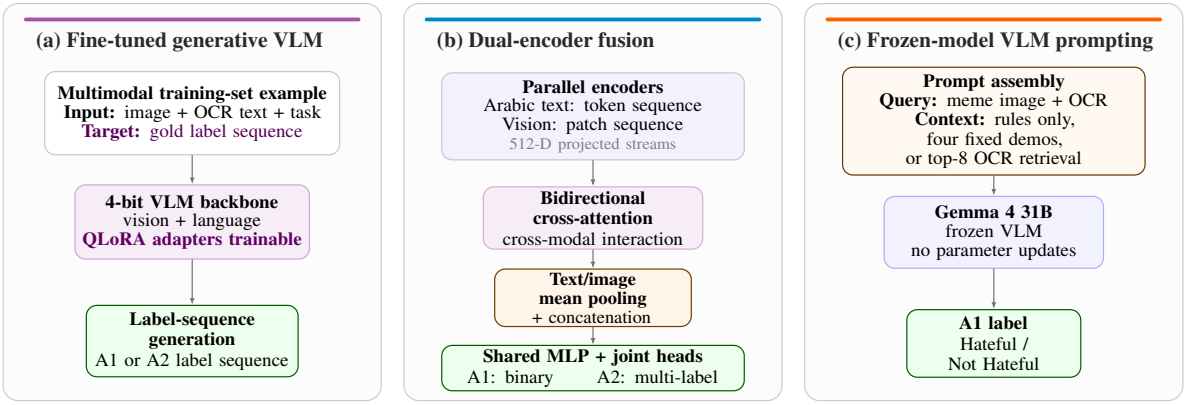


Figure 1: Overview of the three paradigms: (a) a QLoRA-adapted generative VLM; (b) projected Arabic-text and vision streams fused by bidirectional cross-attention; and (c) a frozen VLM prompted with rules, fixed demonstrations, or retrieved demonstrations.

layer-normalization, and feed-forward sublayers (Vaswani et al., 2017). A small shared MLP maps h to z, and the task heads compute $p_{\mathrm{A1}} = \sigma(w^\top z + b)$ and $\mathbf{p}_{\mathrm{A2}} = \sigma(Wz + \mathbf{b})$. They are trained jointly with $\mathcal{L}_{\mathrm{dual}} = \mathrm{BCE}(y_{\mathrm{A1}}, p_{\mathrm{A1}}) + \mathrm{BCE}_{\boldsymbol{\omega}}(\mathbf{y}_{\mathrm{A2}}, \mathbf{p}_{\mathrm{A2}})$, where $\boldsymbol{\omega}$ contains inverse-frequency positive-class weights for A2. The vision encoder remains frozen, while the text encoder, fusion layers, and heads are fine-tuned.

Encoder configurations. We instantiate two encoder configurations. **CLIP–MARBERTv2** pairs MARBERTv2 with CLIP ViT-L/14 and processes images at 224px resolution. **SigLIP–AraBERT-Twitter** pairs AraBERT-Twitter with SigLIP-L/16 and uses a resolution of 384px.

Calibration and ensembling. Because macro-F_1 is sensitive to decision thresholds under class imbalance, we tune a single A1 threshold and a separate threshold for each A2 category on the development set. We also combine the two systems using weighted soft voting over their predicted probabilities. The mixture weight is selected separately for each subtask, after which we recalibrate the corresponding decision thresholds. We fix all selected weights and thresholds before evaluating the test set.

4.3 Frozen-model VLM prompting

We query a frozen Gemma 4 31B model under three exploratory A1 configurations (Gemma Team, 2026). These experiments use *frozen-model prompting*, with model weights held fixed during inference. The ICL configurations use labeled demonstrations and class-prior information, while retrieval also uses A2 label metadata. These experiments were not included in our official submissions for the final evaluation.

Zero-shot prompting provides the meme image, OCR text, a short task definition, and the two allowed labels. *Static multimodal ICL* adds four manually selected demonstrations, balanced between the two A1 classes. Each demonstration contains an image, its OCR text, the gold label, and a brief rationale. *Retrieval-based ICL* replaces this fixed context with the eight nearest training-set examples. We retrieve them by cosine similarity after indexing OCR text and A2 metadata with Snowflake Arctic-Embed-L-v2.0 (Yu et al., 2024), then insert their images, text, and label-derived rationales into the conversation.

We evaluate all three baselines exclusively on the development-test split. The three configurations differ in their system prompts as well as in their contexts, so they are best read as three complete prompting recipes rather than as an ablation of a single component (Appendix C.3). Appendix B.2 provides the complete prompts, context-construction procedure, and inference settings.

5 Experimental Setup

Input processing. For the dual encoders, we remove Arabic diacritics and tatweel, normalize common variants of Alef, Ya, and Hamza, and shorten extended character repetitions. The AraBERT-Twitter branch also applies the model's native preprocessor. We convert all images to RGB and process them with the preprocessing pipeline associated with each vision encoder. For the VLM experiments, each multimodal chat example contains the image, OCR text, task instruction, and task-specific

System / formulation	A1	A2
CLIP–MARBERTv2	0.7374	0.3136*
SigLIP–AraBERT-Twitter	0.7345	**0.3851**
Soft-vote ensemble	0.7443	0.3136
Qwen3-VL	**0.7610***	0.2119
Llama 3.2 Vision	0.6952	0.1832
Gemma 4 E4B	0.4906	0.2418

Table 1: Our main results on the test set for both subtasks. An asterisk marks final official shared-task scores.

target label sequence.

Training and model selection. The dual encoders use AdamW with separate learning rates for the text backbone and newly initialized fusion layers and classification heads. We train separate VLMs for A1 and A2 using 4-bit QLoRA and allow a longer generation budget for A2 to accommodate multiple labels. We use the development set to select model checkpoints, ensemble weights, and decision thresholds. Full experimental settings and implementation details are provided in Appendix A.

Comparison protocol. We evaluate the fine-tuned VLMs and dual encoders on the development set and, where corresponding submission outputs are available, on the test set. The frozen-model prompting baselines were evaluated only on the development-test split and were not submitted for final evaluation.

6 Results and Analysis

Table 1 presents macro-F_1 scores on the test set, which constitute our main comparison. Appendix C reports the development-set results used for model selection, together with the complete A1 metrics for the fine-tuned VLMs. Because the frozen-model prompting configurations were evaluated only on the development-test split, we discuss their results separately in Appendix C.3. These prompting results do not isolate the effect of retrieval: static and retrieval-based ICL differ in demonstration selection and count, system prompts, and whether an explicit chain-of-thought block is required.

On subtask A1, Qwen3-VL records the highest test-set score, with a macro-F_1 of 0.7610. The soft-vote ensemble reaches 0.7443, compared with 0.7374 for CLIP–MARBERTv2 and 0.7345 for SigLIP–AraBERT-Twitter. It also slightly outperforms both component systems on the development set, reaching 0.7756 compared with 0.7613 for CLIP–MARBERTv2 and 0.7663 for SigLIP–AraBERT-Twitter (Appendix C.1). The gains on both splits indicate a modest but consistent benefit from probability-level fusion. Llama 3.2 Vision obtains 0.6952. Gemma 4 E4B scores 0.4906 despite matching Qwen3-VL's development-set macro-F_1 of 0.73. Its test-set accuracy of 0.7220 does not reflect its uneven class-level results: F_1 is 0.1472 for *Hateful* and 0.8339 for *Not Hateful*. The model therefore favors the majority class on the test set, indicating that its development-set performance does not transfer reliably.

On subtask A2, SigLIP–AraBERT-Twitter achieves the highest macro-F_1 at 0.3851, exceeding both CLIP–MARBERTv2 and the soft-vote ensemble. In our setup, the dual-encoder systems outperform the generative VLMs on A2. The dual encoders are trained jointly on A1 and A2, whereas the VLMs are trained separately for each subtask. Appendix C.2 shows that CLIP–MARBERTv2 scores higher on the development set (0.4236 vs. 0.3880), while SigLIP–AraBERT-Twitter scores higher on the test set (0.3851 vs. 0.3136). Because the A2 ensemble weight was selected on the development set, all weight was assigned to CLIP–MARBERTv2. The ensemble therefore matches its test-set score rather than gaining from SigLIP–AraBERT-Twitter. The development-set ranking and corresponding ensemble choice did not transfer to the test set. These results cannot determine whether this difference arises from label sparsity, differences between the splits, or overfitting in threshold selection.

6.1 Error Analysis

We analyze all 125 development-set errors made by Qwen3-VL, our best A1 model on the test set. The 63 false negatives and 62 false positives are nearly balanced. Because hate subtypes are applied only to *Hateful* examples and *Humor* and *Sarcasm* only to *Not Hateful* examples, their separation by error direction follows the dataset convention. More informatively, *Mocking* or *Slurs* is the sole annotation in 37 of the 63 false negatives, 12 false negatives carry two or three hate labels, and a single Arabic term is misclassified in both abusive and non-targeted contexts. These cases point to difficulty resolving target, intent, and contextual meaning. Appendix D provides the full breakdown with illustrative examples.

7 Conclusion

We compare three approaches to Arabic hateful meme detection: QLoRA adaptation of Qwen3-VL, Llama 3.2 Vision, and Gemma 4 E4B; cross-attention fusion with CLIP–MARBERTv2 and SigLIP–AraBERT-Twitter, including soft-vote ensembling; and frozen Gemma 4 31B prompting with zero-shot, static ICL, and retrieval-based ICL configurations. Qwen3-VL achieves the highest test-set macro-F_1 on A1, while SigLIP–AraBERT-Twitter leads A2. Static ICL scores highest among the prompting configurations on the dev-test split. The error analysis highlights challenges involving target and stance resolution, cultural context, non-literal humor, and OCR quality. Future work will explore stronger multimodal retrieval, prompting, and agent-based approaches.

Limitations

Some experiments use different evaluation splits and therefore do not support controlled cross-model comparisons. Most fine-tuned systems are evaluated with a single random seed, and hardware constraints required different batch sizes across models. The dual encoders freeze their vision towers, whereas QLoRA adapts both modalities, so comparisons also reflect different training budgets.

Although our dual encoders use loss reweighting and per-label threshold calibration, we do not systematically compare imbalance-handling strategies under a fixed computational budget. Future work should compare reweighting, resampling, and calibration, including their interaction with parameter-efficient adaptation. Without a per-label A2 breakdown, we cannot determine whether SigLIP–AraBERT-Twitter's higher macro-F_1 reflects broad improvements across labels or gains concentrated in a few categories.

The frozen-model configurations were evaluated only on the development-test split and are therefore not directly comparable with systems evaluated on the test split; they also differ from one another in their system prompts as well as their contexts, so they compare whole prompting recipes rather than isolating any single factor. Finally, our official submissions were Qwen3-VL for A1 and CLIP–MARBERTv2 for A2, so the strongest A2 result we report (SigLIP–AraBERT-Twitter) was identified after the submission deadline and did not contribute to our official ranking.

Ethical Considerations

This work involves hateful and potentially offensive memes, which may expose annotators and researchers to harmful content. Automated predictions can also reproduce social, cultural, or dialectal biases: false positives may suppress benign humor or political expression, whereas false negatives may leave harmful content unflagged. These systems should therefore support, rather than replace, culturally informed human review in consequential moderation settings. We use the shared-task data only for research under its access terms.

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A Experimental and Implementation Details

We train the dual encoders with AdamW, using a learning rate of 2×10^{-5} for the text encoder and 1×10^{-4} for the fusion layers and classification heads. We apply a weight decay of 0.01 and a cosine learning-rate schedule with a 10% warmup period. The batch size is 16, the maximum text length is 96 tokens, and the shared projection dimension is 512. The fusion module uses eight attention heads, with dropout rates of 0.1 in the fusion layers and 0.2 in the task heads. Training uses FP16 mixed precision. Character runs of length three or more are reduced to two, and the AraBERT-Twitter branch applies its emoji-aware native preprocessor. For A2, we cap the inverse-frequency positive class weights $\boldsymbol{\omega}$ at 10. We search decision thresholds from 0.05 to 0.95 in increments of 0.01 and search the ensemble simplex in increments of 0.1.

The VLM experiments use 4-bit NF4 backbones, zero LoRA dropout, no bias terms, Unsloth gradient checkpointing, and a maximum sequence length of 2048. For A1, we set both the LoRA rank and α to 16 and train for three epochs with an effective batch size of 8, a learning rate of 2×10^{-4}, a linear schedule with five warmup steps, weight decay of 0.001, and an 8-bit AdamW optimizer. The dedicated A2 runs use the same settings, except that the LoRA rank and α are 32 and generation allows up to 24 new tokens. Because of memory constraints, Gemma 4 E4B uses an effective batch size of 4. All runs use a random seed of 3407.

For Gemma 4, we use the `unsloth/gemma-4-E4B-it` checkpoint with its non-thinking chat template, which places the image before the text. We apply LoRA to all linear layers and keep key–value caching enabled during generation.

A.1 Frozen-Model Inference and Retrieval Settings

All three exploratory A1 configurations use the frozen Gemma 4 31B model through OpenRouter, with a temperature of 0 and a maximum of 64 generated tokens. Each prompt requests a final prediction of `Hateful` or `Not Hateful`. OpenRouter's separate reasoning mode is disabled; the explicit `<thought>` block requested by the retrieval configuration is part of the ordinary generated response rather than provider-side reasoning output.

Static ICL uses four fixed training-set examples, balanced between the two A1 classes. Retrieval-based ICL selects the eight most similar training-set examples according to cosine similarity. We index each training-set item by concatenating its OCR text with its fine-grained label metadata, whereas the query representation contains only the OCR text of the current meme. Snowflake Arctic-Embed-L-v2.0 encodes both representations. We then insert the retrieved images, OCR text, and label-derived rationales as alternating user and assistant turns. All three configurations are evaluated on the development-test split using macro-F_1 and were not submitted for final evaluation. Appendix B.2 reproduces the exact prompts and response formats.

A.2 Model and API Resources

Table 2 lists the exact public checkpoints and API resources used in the experiments.

Component	Checkpoint or API resource
Qwen3-VL-8B-Instruct (Unsloth 4-bit)	Hugging Face checkpoint
Llama-3.2-11B-Vision-Instruct (Unsloth 4-bit)	Hugging Face checkpoint
Gemma 4 E4B	Hugging Face checkpoint
Gemma 4 31B	OpenRouter model page
MARBERTv2	Hugging Face model card
AraBERT-Twitter	Hugging Face model card
CLIP ViT-L/14	Hugging Face model card
SigLIP-L/16 (384 px)	Hugging Face model card
Snowflake Arctic-Embed-L-v2.0	Hugging Face model card

Table 2: Public model checkpoints and API resources used in the experiments.

B Prompt Templates and Inference Configurations

B.1 Fine-tuning Prompts and Output Formats

The following instructions were used in the VLM fine-tuning pipelines. The chat template supplies the image and OCR text separately.

Separate A1 Binary Generation

```
Detect the classification of hateful Arabic memes from image and extracted text. The classification
labels are only ['Hateful', 'Not Hateful'], return only the label.
```

Dedicated A2 Multi-label Generation

```
Detect the fine-grained classification(s) of hateful Arabic memes from the image and extracted
text. A meme can have one or more labels at the same time. The classification labels are only
[Contempt, Dehumanization, Exclusion, Humor, Incitement, Inferiority, Mocking, Other, Sarcasm,
Slurs]. Return only the label(s) that apply, separated by a comma if there is more than one, and
nothing else.
```

Generation is deterministic. A1 uses at most 8 new tokens, while A2 uses up to 24 to accommodate multiple labels.

B.2 Frozen-Model Prompting Configurations

This appendix reproduces the exact system and user prompts used by the Gemma 4 31B zero-shot and in-context baselines through OpenRouter. At inference time, text enclosed by «. . . » is replaced with the OCR transcript of the current meme. All configurations use the frozen model with a temperature of 0 and a maximum of 64 generated tokens. They are evaluated only on the development-test split and were not submitted for final evaluation.

B.2.1 Culturally-Aware Zero-Shot Baseline

System Prompt

```
You are an expert Arabic content moderation specialist focused on detecting hate speech in social
media memes.
A meme consists of an image combined with overlaid Arabic text. Hate speech in memes is often:
- Non-compositional: the image and text are harmless alone but hateful together.
- Implicit: conveyed through cultural references, sarcasm, or dehumanizing metaphors.
- Dialectal: written in Egyptian, Syrian, Palestinian, or other Arabic dialects.
Classification rules:
"Hateful" — The meme targets, demeans, incites hatred against, or dehumanizes a person or group.
"Not Hateful" — The meme does not contain hate speech.
Respond, verbatim and ONLY, with EXACTLY ONE of: Hateful OR Not Hateful
```

User Prompt Template

```
OCR-extracted text from the meme: <<{ocr_text}>>
Analyze the meme image and the text together. Is this meme hateful?
Respond with exactly: Hateful OR Not Hateful
```

B.2.2 Static Multimodal In-Context Learning

System Prompt

```
You are an expert Arabic-language content moderator specializing in hateful meme detection.

CRITICAL OUTPUT FORMAT REQUIREMENT:

Your response MUST start on LINE 1 with EXACTLY one of these two labels:

Hateful

Not Hateful

You may provide a brief 1-2 sentence explanation after the label.

IMPORTANT CALIBRATION: In this dataset, approximately 38% of memes ARE hateful. Your threshold must reflect this — do NOT default to Not Hateful when uncertain.

Hateful memes in Arabic often use:

1. Dehumanization — comparing people to animals, vermin, or objects (e.g. 'كلب', 'قرد', 'بهيمة', 'حشرة')

2. Incitement — calls for violence, expulsion, or harm (e.g. 'اقتلوهم', 'ارحلوا', 'نظفوا البلد منهم')

3. Slurs & derogatory dialect — Egyptian, Syrian, Palestinian, Gulf slang that demeans ethnicity, religion, or gender

4. Religious mockery — belittling Islamic, Christian, or Jewish practices or symbols

5. Implicit contempt — images where the visual framing (e.g. an animal next to a political figure, or a degrading visual metaphor) carries the hate even without explicit words

6. Non-compositional hate — the image alone or text alone seems fine, but together they demean a group

Decision rule:

- If there is ANY credible indication of hate (dehumanization, slurs, incitement, mockery of a group, exclusion) — label it Hateful.

- Only label Not Hateful when you are confident the meme contains no targeting of any group.
```

Context construction. The user template below was preceded by four fixed training-set examples: two hateful and two non-hateful. Each example contained the image, OCR text, and an assistant response with the gold label and a short rationale.

User Prompt Template

```
OCR-extracted text from the meme: <<{ocr_text}>>

Analyze the image and text together. Is this meme hateful?

REMINDER: Output EXACTLY 'Hateful' or 'Not Hateful' on Line 1.
```

B.2.3 Dynamic Retrieval-Augmented ICL

System Prompt

```
You are an expert Arabic-language content moderator specializing in hateful meme detection.
CRITICAL OUTPUT FORMAT REQUIREMENT:
You MUST use Chain-of-Thought reasoning to evaluate the meme first.
Write your reasoning inside <thought>...</thought> XML tags.
Then, on a NEW LINE after the thought block, output EXACTLY one of these two labels:
Hateful
Not Hateful
IMPORTANT CALIBRATION: In this dataset, approximately 38% of memes ARE hateful. Your threshold
must reflect this. If there is ANY indication of hate, label it Hateful.
Hateful memes in Arabic often use:
1. Dehumanization — comparing people to animals, vermin, or objects
2. Incitement — calls for violence, expulsion, or harm
3. Slurs & derogatory dialect — slang that demeans ethnicity, religion, or gender
4. Religious mockery — belittling Islamic, Christian, or Jewish practices or symbols
5. Implicit contempt — degrading visual metaphors or juxtapositions
6. Non-compositional hate — image + text together are hateful even if each alone is not
Decision rule:
- If there is ANY credible indication of hate — label it Hateful.
- Only label Not Hateful when you are confident the meme is completely harmless.
```

Context construction. The user template below was preceded by the eight nearest retrieved examples, inserted as alternating user and assistant turns. These demonstrations included the image, OCR text, and the label-derived rationale used to establish the reasoning pattern.

User Prompt Template

```
OCR-extracted text from the meme: <<{ocr_text}>>
Analyze the image and text together. Is this meme hateful?
Use <thought> tags first, then output EXACTLY 'Hateful' or 'Not Hateful'.
```

C Additional Results

C.1 Detailed A1 Metrics

Table 3 reports the complete A1 metrics recorded for the three fine-tuned VLMs. We present results for both the development set and the test set.

Model	Split	Accuracy	Macro-F_1	Macro-P	Macro-R	Weighted-F_1	F_1 (Hateful)	F_1 (Not Hateful)
Qwen3-VL	Development set	0.75	0.73	0.73	0.73	0.75	0.67	0.80
	Test set	0.7920	0.7610	0.7530	0.7740	0.7961	0.6750	0.8471
Llama 3.2 Vision	Development set	0.73	0.69	0.73	0.68	0.72	0.58	0.80
	Test set	0.7580	0.6952	0.7080	0.6872	0.7516	0.5568	0.8336
Gemma 4 E4B	Development set	0.75	0.73	0.73	0.73	0.75	0.67	0.80
	Test set	0.7220	0.4906	0.7598	0.5363	0.6307	0.1472	0.8339

Table 3: Detailed A1 results for the fine-tuned generative VLMs.

C.2 Dual-Encoder Fusion Results on the Development Set

Table 4 reports the development-set scores used to select the decision thresholds and ensemble weights for the dual-encoder systems.

System	A1 macro-F_1	A2 macro-F_1
CLIP–MARBERTv2	0.7613	0.4236
SigLIP–AraBERT-Twitter	0.7663	0.3880
Soft-vote	0.7756	0.4236

Table 4: Dual-encoder fusion results on the development set.

C.3 Frozen-Model Prompting Results on the Development-Test Set

Table 5 compares the three Gemma 4 31B prompting configurations on the development-test split. These exploratory experiments were not submitted for final evaluation.

Configuration	Prompt context	k	Explicit CoT	Macro-F_1
Culturally-aware zero-shot	Rules only	—	No	0.5951
Static multimodal ICL	Four fixed examples	—	No	**0.6851**
Retrieval-based ICL	Semantic retrieval	8	Yes	0.6717

Table 5: Frozen-model prompting results with Gemma 4 31B for A1 on the development-test split.

Static ICL improves macro-F_1 by 0.0900 over the culturally-aware zero-shot prompt, and retrieval-based ICL improves by 0.0766 while remaining 0.0134 below the static configuration.

These gaps should not be read as isolating the contribution of the demonstrations. As Appendix B.2 shows, both ICL system prompts include an explicit class-prior instruction stating that roughly 38% of the dataset is hateful and a six-category inventory of Arabic hate patterns; the static prompt also includes lexical examples. Their output-format requirements and decision rules differ from those of the zero-shot prompt. The static and retrieval configurations differ in turn along three axes simultaneously: fixed versus retrieved demonstrations, four versus eight of them, and the absence versus presence of a mandatory chain-of-thought block. We therefore report these numbers as an exploratory comparison of three complete prompting recipes and do not attribute the differences to demonstration selection or retrieval quality. The class-prior instruction is derived from the labeled training split, so the ICL configurations still rely on labeled data despite keeping model weights fixed.

D Detailed Error Analysis

We audit all 125 errors made by Qwen3-VL on the development set. False negatives account for 63 cases (50.4%; Wilson 95% CI: 41.8–59.0%), while 62 are false positives. The near-even balance does

Annotation pattern	FN	FP
Mocking	22	—
Slurs	15	—
Incitement	7	—
Dehumanization	5	—
Exclusion	1	—
Multi-label hate patterns	12	—
Sarcasm	—	27
Humor	—	20
Other	1	15
Total	63	62

Table 6: Qwen3-VL errors on the development set, grouped by error direction and fine-grained annotation pattern. The multi-label row aggregates all false negatives carrying two or three hate-subtype labels.

not indicate a general tendency to under- or over-flag. The main difference appears in the fine-grained annotations shown in Table 6.

By task convention, hateful examples receive hate-subtype labels, whereas non-hateful examples receive *Humor* or *Sarcasm*; *Other* can occur with either class. Their separation by error direction in Table 6 therefore follows the annotation design rather than indicating an independent model tendency. Within the false negatives, *Mocking* or *Slurs* is the only annotation in 37 cases; both tags also occur inside the multi-label group, so the number of false negatives containing either one is higher. Twelve of 63 (19%) carry two or three hate-subtype labels. No false positive has more than one fine-grained tag.

D.1 Lexical Context

The term الكلب (*dog*) provides a useful example of context sensitivity. Qwen3-VL misses one use in which the word directly insults a person, but flags other cases where it denotes a carpentry clamp or appears in a non-targeted exclamation. The same lexical item therefore produces errors in both directions. This is consistent with an unstable reliance on lexical cues when the model must resolve word sense, reference, and whether a person or group is being targeted.

D.2 Visual Context and OCR Quality

One false negative annotated with *Other* has innocuous OCR text, so the textual input alone does not explain the hateful annotation and the image may provide relevant context. Some examined examples also contain clear OCR corruption. These observations do not establish causality, but they caution against treating the OCR transcript as a complete representation of each meme or attributing every error solely to the classifier's semantic reasoning.

D.3 Illustrative Cases

We select six diagnostic examples that represent complementary error types rather than choosing cases for their visual salience. Figures 2–4 show the original memes, while Table 7 records their gold annotations and model predictions. The English glosses below translate the salient content rather than the noisy OCR strings verbatim.

Panel	Gold annotation	Prediction
(a)	Hateful: Mocking	Not Hateful
(b)	Not Hateful: Sarcasm	Hateful
(c)	Hateful: Incitement, Slurs	Not Hateful
(d)	Hateful: Mocking	Not Hateful
(e)	Not Hateful: Humor	Hateful
(f)	Not Hateful: Humor	Hateful

Table 7: Gold annotations and predictions for the illustrative Qwen3-VL errors on the development set. The category after the colon is the dataset's fine-grained annotation.

Target and stance. Panels (a) and (b) form a contrastive pair around the same lexical family, *fallah* ("peasant"). In (a), the plural term describes the people at a destination and is annotated as group-directed mockery, but Qwen3-VL predicts *Not Hateful*. In (b), the feminine singular is used self-referentially in a visual joke (approximately, "So I am the peasant one?"), yet the model predicts *Hateful*. The reversal shows why the presence of a socially marked term is insufficient without resolving who is targeted and whether the stance is outwardly derogatory or self-directed.



(a) Group-directed use: false negative **(b) Self-referential use: false positive**

Figure 2: A contrastive lexical pair. Qwen3-VL misses the outwardly mocking use but over-flags the self-referential joke, indicating sensitivity to the word without stable resolution of its target and pragmatic role.

Compounded and culturally grounded hate. Panel (c) contains an escalating colloquial exchange: after repeatedly denying jealousy, the final turn threatens to pull another woman by her hair and adds a direct slur. Qwen3-VL nevertheless predicts *Not Hateful* despite the two gold categories and several mutually reinforcing cues. Panel (d) describes a classmate who repeatedly looks toward the narrator with a particular facial expression, calling it "Ali Rabie syndrome." The reaction image supplies the comparison to a locally recognizable comic persona and clarifies what the caption means by "a face like this." Interpreting the dataset's *Mocking* annotation therefore requires connecting the colloquial description to the culturally familiar visual template rather than detecting an explicit insult. Qwen3-VL instead predicts *Not Hateful*.



(c) Compounded incitement and slur **(d) Culturally situated visual comparison**

Figure 3: Two false negatives with different contextual demands. Panel (c) requires following the escalation across a dialogue and retaining multiple applicable labels. Panel (d) requires connecting a colloquial comparison to a locally recognizable reaction image.

Nonliteral humor and noisy OCR. Panel (e) says, approximately, "I hope the Turks occupy us so that I can marry a Turkish man." The occupation scenario is a comic romantic exaggeration rather than a call for harm or an attack on a target, but the model predicts *Hateful*, consistent with an overly literal

reading of politically charged language. Panel (f) is an ordinary family joke about a mother breaking a glass and immediately minimizing the accident. Its supplied OCR contains extensive errors even though the image text is benign. Because the model receives both the image and the OCR transcript, this case cannot establish which modality caused the false positive; it instead shows why input quality complicates causal attribution in multimodal error analysis.



(e) Hyperbolic humor



(f) Benign humor with corrupted OCR

Figure 4: Two false positives. The first tests nonliteral interpretation; the second highlights the difficulty of separating semantic model errors from degradation in an auxiliary OCR input.

Cross-case synthesis. The six cases do not reduce to a single missing lexicon or modality. The contrastive pair requires target and stance resolution; the false negatives require dialogue-level accumulation or a culturally grounded visual comparison; and the false positives require calibration against nonliteral language and uncertain OCR. These cases motivate target-aware pragmatic modeling and explicit treatment of OCR reliability in addition to stronger visual–textual fusion.